



LESSON PLAN

Basic Urban Farming with Hydroponics

WSQ Course Code: TGS-2025053916

Version 2.0 · 29 August 2026

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Document Version Control Record

Version	Date	Description	Approved by
1.0	24 March 2025	Legacy approved course package baseline.	Course Development Team
2.0	29 August 2026	Rebuilt lesson plan aligned to 8-hour delivery flow, v2.0 deck and activity evidence.	Dr Alfred Ang

Table of Contents

- Course Information
- Delivery Schedule
- Facilitation and Assessment Notes

Course Information

Course title	Basic Urban Farming with Hydroponics
Course code	TGS-2025053916
Duration	1 day · 8 hours excluding lunch
TSC	AGR-FOP-2002-1.1 Urban Farming Implementation and Management Control
Assessment	WA-SAQ 30 minutes; PP 30 minutes
Venue	MEOD Farm, 13 Neo Tiew Harvest Lane, Singapore 719020 (verify joining instructions)

Delivery Schedule

Time	Duration	Topic / slides	Facilitation	Learner evidence	Resources
09:00–09:30	0.5 h	Briefing, profiles, access, outcomes and safety · slides 1–17	Questioning; access and equipment-risk examples	Verbal checks	Deck, LG
09:30–10:30	1.0 h	Foundations and systems · slides 18–45	Mechanism diagrams; system comparison	System-selection rationale	Deck, legacy visuals
10:30–10:45	0.25 h	Break	—	—	—
10:45–12:30	1.75 h	Water, nutrients, crop controls and Singapore cases · slides 46–96	Data interpretation; Singapore cases; root diagnosis	pH/EC reasoning; case critique	Deck, meters, LG
12:30–13:30	—	Lunch (not counted)	—	—	—
13:30–14:30	1.0 h	Farm tour · slides 97–106	Guided observation; C/O/I evidence coaching	Completed farm-tour checklist	Activity 01 PDF
14:30–16:00	1.5 h	Hydroponics kit practice · slides 107–118	Demonstrate then coach against SOP; observe A1–A4	Commissioning and correction record	Kit, PPE, meters, Activity 02 PDF
16:00–16:15	0.25 h	Break	—	—	—
16:15–17:00	0.75 h	Aquaponics theory · slides 119–131	Nitrogen-cycle loop; coupled-risk diagnosis	Aquaponics fault explanation	Deck, LG
17:00–18:00	1.0 h	Assessment and close · slides 132–139	Administer WA-SAQ then PP; feedback	Candidate responses and assessor observation	Assessment papers

Final slide ranges are subject to the generated deck inventory (139 slides); the trainer should use the section dividers and slide-map JSON for exact navigation.

Facilitation and Assessment Notes

- Do not present the detailed operating procedure from slides; use the Learner Guide and activity PDFs at the workbench.
- Before the farm tour, confirm site rules, photography permissions, emergency arrangements and learner accessibility needs.
- Use questioning to test why each control exists, not only whether a learner can repeat a setpoint.
- For practical performance, observe safe preparation, operation to target, compliance/housekeeping and cleaning/maintenance.
- Stop any task when equipment, chemical, electrical or hygiene controls are missing.